

# Complete Summary of Results in Chapters 4–7

## Chapter 4: Measurable Functions

- **Theorem 4.7:**  $T$  is  $(\mathcal{A}, \mathcal{A}')$ -measurable  $\iff T^{-1}(\mathcal{E}') \subseteq \mathcal{A}$ .
- **Proposition 4.8:**  $f : \Omega \rightarrow \mathbb{R}$  is measurable  $\iff \{f \leq x\} \in \mathcal{A}$  for all  $x$ , and several equivalent formulations.
- **Proposition 4.9:** If  $f, g$  are measurable, then the sets  $\{f < g\}, \{f \leq g\}, \{f = g\}, \{f \neq g\}$  are measurable.
- **Theorem 4.10:** If  $f_n$  is a sequence of measurable functions, then  $\sup f_n$ ,  $\inf f_n$ ,  $\limsup f_n$ , and  $\liminf f_n$  are measurable.
- **Theorem 4.11:** If  $f, g$  are measurable, then so are  $f + g$ ,  $f - g$ ,  $fg$ , and  $f/g$  (where defined).

## Chapter 5: Lebesgue Integral

- **Theorem 5.14:** If  $f \geq 0$  is measurable, and  $f = \sum_n f_n$  with  $f_n \geq 0$ , then  $\int f = \sum_n \int f_n$ .
- **Theorem 5.15:** If  $f = g$   $\mu$ -a.e. and  $f$  is integrable, then  $g$  is integrable and  $\int f = \int g$ .
- **Corollary 5.16:** If  $N \in \mathcal{A}$  with  $\mu(N) = 0$ , then  $\int_N f d\mu = 0$ .
- **Theorem 5.17:** If  $f = g$   $\mu$ -a.e. and  $f$  is integrable, then  $\int f = \int g$ .

## Chapter 6: Convergence Theorems

- **Theorem 6.1 (Fatou's Lemma):**  $\int \liminf f_n \leq \liminf \int f_n$  for  $f_n \geq 0$  measurable.
- **Theorem 6.2 (Monotone Convergence):** If  $f_n \uparrow f$  and all  $f_n \geq 0$ , then  $\int f_n \rightarrow \int f$ .
- **Theorem 6.3 (Dominated Convergence):** If  $f_n \rightarrow f$  a.e.,  $|f_n| \leq g$  and  $g$  integrable, then  $\int f_n \rightarrow \int f$ .

## Chapter 7: Product Measures and Fubini's Theorem

- **Theorem 7.7:** If  $\mathcal{R}$  is a semiring closed under intersection, then  $\sigma(\mathcal{R}) = \delta(\mathcal{R})$ .
- **Theorem 7.9:** If  $A \in \mathcal{A}_1 \otimes \mathcal{A}_2$ , then sections  $A(\omega_1, \cdot)$  and  $A(\cdot, \omega_2)$  are measurable.
- **Theorem 7.10:** Defines product measure using slices and shows  $\mu$  is a measure on the product  $\sigma$ -algebra.
- **Theorem 7.11:** If  $\mu = \nu$  on a generator closed under intersection, and total mass is equal, then  $\mu = \nu$  on  $\sigma$ -algebra.
- **Fubini's Theorem:** If  $f$  is integrable on product space, then iterated integrals equal the product integral.